Linear Perturbations and Structure Growth

From Projection Signatures to Effective Growth Dynamics
in Structural-Selection Cosmology

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Abstract

Previous MAAT cosmology papers introduced structural projection observables and growth-signature proxies, but the growth sector remained limited to template-level or growth-index estimates. This paper introduces the first explicit linear-growth equation in the MAAT structural-selection cosmology sequence.

The construction embeds a bounded projection-derived structural coupling into the standard sub-horizon matter-growth equation through

$$\frac{G_{\rm eff}(z)}{G} = \mu(z) = 1 + \eta\,\widehat{\mathcal{C}}_{\rm proj}(z),$$

where $\widehat{\mathcal{C}}_{\rm proj}(z)$ is the bounded projection kernel derived from the response-based projection observable and η controls the maximum effective growth-coupling deformation over the tested interval $0 \le z \le 3$.

Numerically, we scan $\eta \in [0, 0.08]$ using 41 grid points and compare the resulting $f\sigma_8(z)$ curves with a compact set of 13 growth measurements. All tested branches remain stable in the effective benchmark. The best value in this compact comparison is $\eta = 0$, with $\chi^2_{\Lambda\rm CDM} = 12.2021$. A representative nonzero branch, $\eta = 0.02$, gives

$$\max_{z\le 3} |\Delta D/D| = 0.7656\%, \qquad \max_{z\le 3} |\Delta f\sigma_8/f\sigma_8| = 0.5441\%,$$

with $\max|\mu - 1| = 1.9966\%$ and $\chi^2 = 12.5737$.

The result is not evidence for modified gravity and not a precision cosmological fit. Its contribution is methodological: MAAT projection structure can be inserted into an explicit linear perturbation equation without generating large or unstable growth effects.

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1 Purpose and Status

The recent MAAT cosmology sequence moved through three stages:

- projection diagnostics for expansion and structure,
- observable growth-signature templates,
- blind residual-structure tests.

Those papers established that the projection observable is not merely a relabelled background quantity, but they did not yet solve a perturbation equation.

The purpose of Paper 43 is therefore narrow:

> *Can a projection-derived MAAT structural coupling be inserted into an explicit linear matter-growth equation while remaining perturbatively controlled and stable?*

This paper is an effective linear-growth benchmark. It is not a Boltzmann-code calculation, not a CMB or weak-lensing likelihood, and not a claim of observational preference over ΛCDM. It is the first bridge from projection signatures to perturbation dynamics.

2 Background Cosmology

We use a spatially flat matter-plus-Λ reference background with

$$E^2(a) = \frac{H^2(a)}{H_0^2} = \Omega_{m0} a^{-3} + \Omega_{\Lambda 0}. \tag{1}$$

The numerical benchmark uses Planck-normalised reference values

$$H_0 = 67.4, \qquad \Omega_{m0} = 0.315, \qquad \Omega_{\Lambda 0} = 0.685, \qquad \sigma_{8,0} = 0.811. \tag{2}$$

The matter fraction at scale factor a is

$$\Omega_m(a) = \frac{\Omega_{m0} a^{-3}}{E^2(a)}. \tag{3}$$

For this background,

$$\frac{\mathrm{d}\ln H}{\mathrm{d}\ln a} = -\frac{3}{2}\Omega_m(a). \tag{4}$$

3 Projection-Derived Coupling

Paper 42 derived the projection shape from response-based sector weights. The raw projection observable $\mathcal{C}_{\rm proj}(z)$ grows strongly with redshift. For a perturbative growth coupling this raw quantity is too large to insert directly. We therefore use only its bounded shape over the tested interval:

$$\boxed{\widehat{\mathcal{C}}_{\rm proj}(z) = \frac{\mathcal{C}_{\rm proj}(z) - \mathcal{C}_{\rm proj}(0)}{\mathcal{C}_{\rm proj}(z_{\rm max}) - \mathcal{C}_{\rm proj}(0)}, \qquad z_{\rm max} = 3.} \tag{5}$$

By construction,

$$0 \le \widehat{\mathcal{C}}_{\rm proj}(z) \le 1 \qquad (0 \le z \le 3). \tag{6}$$

The effective perturbative coupling is defined as

$$\boxed{\mu(z) = \frac{G_{\rm eff}(z)}{G} = 1 + \eta\,\widehat{\mathcal{C}}_{\rm proj}(z).} \tag{7}$$

Thus η has a simple operational meaning: it is the maximum fractional modification of the effective growth coupling over the benchmark interval.

This should not be interpreted as a fundamental variation of Newton's constant. It is an effective structural response term inserted into the linear growth sector.

4 Linear Growth Equation

For sub-horizon scalar matter perturbations, the standard linear growth equation is

$$\ddot{\delta} + 2H\dot{\delta} - 4\pi G\rho_m\delta = 0. \tag{8}$$

The MAAT benchmark replaces G by the effective response $G_{\rm eff} = G\mu(z)$:

$$\ddot{\delta} + 2H\dot{\delta} - 4\pi G\mu(z)\rho_m\delta = 0. \tag{9}$$

Using $N = \ln a$ as evolution variable gives

$$\boxed{\frac{\mathrm{d}^2\delta}{\mathrm{d}N^2} + \left[2 + \frac{\mathrm{d}\ln H}{\mathrm{d}N}\right]\frac{\mathrm{d}\delta}{\mathrm{d}N} - \frac{3}{2}\Omega_m(a)\mu(a)\delta = 0.} \tag{10}$$

For $\eta = 0$, Eq. (10) reduces exactly to the reference ΛCDM growth equation in the same background.

Initial conditions are imposed in the matter-dominated regime:

$$a_{\rm ini} = 0.01, \qquad \delta(a_{\rm ini}) = a_{\rm ini}, \qquad \frac{\mathrm{d}\delta}{\mathrm{d}N}(a_{\rm ini}) = a_{\rm ini}. \tag{11}$$

5 Growth Observables

The linear growth factor is normalised to the present epoch:

$$D(a) = \frac{\delta(a)}{\delta(a=1)}. \tag{12}$$

The logarithmic growth rate is

$$f(a) = \frac{\mathrm{d}\ln D}{\mathrm{d}\ln a} = \frac{\mathrm{d}\ln\delta}{\mathrm{d}N}. \tag{13}$$

The observable growth combination is

$$f\sigma_8(a) = f(a)\,D(a)\,\sigma_{8,0}. \tag{14}$$

Relative deviations are reported as

$$\frac{\Delta f\sigma_8}{f\sigma_8} = \frac{(f\sigma_8)_{\rm MAAT} - (f\sigma_8)_{\Lambda\rm CDM}}{(f\sigma_8)_{\Lambda\rm CDM}}. \tag{15}$$

6 Numerical Benchmark

The code scans

$$\eta \in [0, 0.08] \tag{16}$$

with 41 uniformly spaced grid points. For each value, Eq. (10) is integrated with a fourth-order Runge–Kutta scheme using 2600 steps in $N = \ln a$.

The benchmark checks basic effective stability:

$$\mu(z) > 0, \qquad D(z) > 0, \qquad f(z) > 0 \tag{17}$$

over the tested range.

The compact growth comparison set contains 13 $f\sigma_8$ points. The diagnostic chi-square is

$$\chi^2(\eta) = \sum_i \left[\frac{(f\sigma_8)_{\rm obs}(z_i) - (f\sigma_8)_{\rm model}(z_i;\eta)}{\sigma_i} \right]^2 . \tag{18}$$

No covariance matrix, survey likelihood, nuisance modelling, or MCMC parameter inference is used.

7 Results

7.1 Eta Scan

Table 1: Paper 43 eta scan summary.

Quantity	**Result**
Eta range	$[0, 0.08]$
Eta grid points	41
Stable branches	41/41
Growth comparison points	13
Best eta	0.0000
$\chi^2_{\Lambda\rm CDM}$	12.2021
Best χ^2	12.2021

The compact growth comparison does not prefer a nonzero MAAT coupling. The best point is the ΛCDM limit $\eta = 0$. This is scientifically useful: the first perturbation-level insertion is not immediately excluded by instability, but it is also not claimed as a detected signal.

7.2 Representative Nonzero Branch

For the representative branch $\eta = 0.02$:

Table 2: Representative nonzero MAAT growth branch.

Quantity	**Value**
η	0.0200
$\max_{z\le 3} \lvert\Delta D/D\rvert$	0.7656%
$\max_{z\le 3} \lvert\Delta f\sigma_8/f\sigma_8\rvert$	0.5441%
$\max_{z\le 3} \lvert\mu - 1\rvert$	1.9966%
χ^2	12.5737

The deformation remains perturbatively small. The effective gravitational response is positive throughout the tested range, and both D and f remain positive.

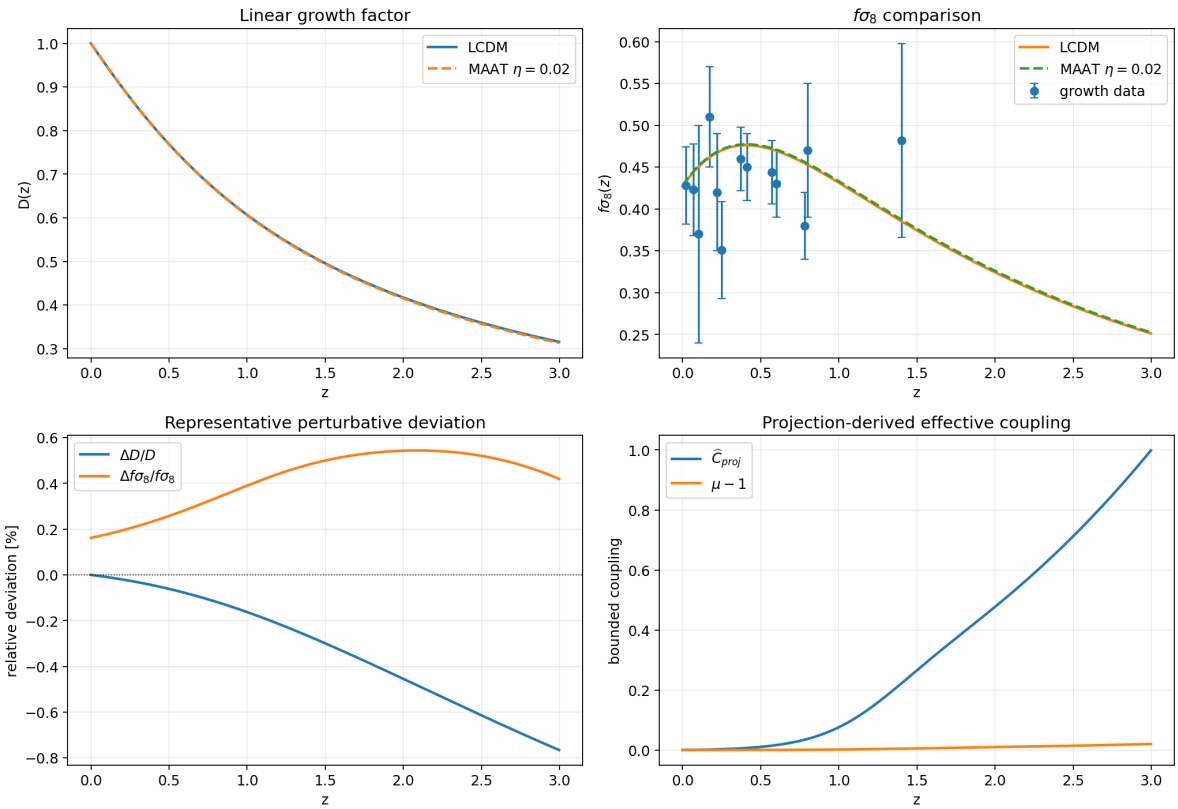

Figure 1: Paper 43 linear-growth perturbation summary. Top left: growth factor comparison. Top right: $f\sigma_8$ comparison with the compact growth-data table. Bottom left: relative deviations for the representative branch $\eta = 0.02$. Bottom right: bounded projection kernel and induced effective coupling.

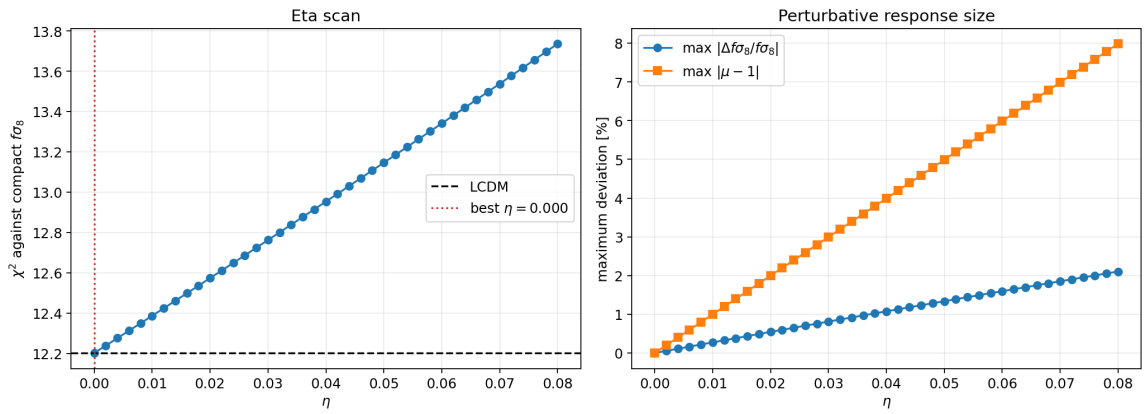

Figure 2: Eta scan. Left: χ^2 against the compact $f\sigma_8$ comparison set. Right: maximum perturbative response size as a function of η. The best value is $\eta = 0$, while all tested nonzero branches remain stable.

8 Interpretation

The key result is not an improvement over ΛCDM. Rather, Paper 43 shows that the projection layer can be promoted from a template-level observable into an explicit linear-growth equation.

The structural chain is

$$\mathcal{C}_{\mathrm{proj}}(z) \to \widehat{\mathcal{C}}_{\mathrm{proj}}(z) \to \mu(z) \to D(z),\ f(z),\ f\sigma_8(z). \tag{19}$$

This is the first MAAT cosmology paper in which the projection observable modifies the growth dynamics through a differential equation rather than through a direct signature template.

The result is conservative:

- nonzero η produces controlled percent-level coupling changes,
- the representative observable deviation remains below 1%,
- the compact comparison set prefers $\eta = 0$,
- no tested branch shows a basic effective-growth instability.

Thus MAAT is not favoured over ΛCDM by this benchmark, but the perturbation-level route is viable enough to motivate a full Boltzmann-level implementation.

9 Limitations

- The growth equation is an effective sub-horizon scalar-growth benchmark, not a full relativistic perturbation system.
- The calculation has no metric perturbations, no CMB anisotropies, no weak-lensing kernels, and no gauge-consistency analysis.
- The compact $f\sigma_8$ table is treated with diagonal errors only.
- No survey covariance matrix or full likelihood is used.
- The background cosmology is fixed to a Planck-normalised ΛCDM reference.
- The projection kernel is bounded over $0 \leq z \leq 3$ for perturbative control; this is an operational benchmark convention, not a first-principles normalisation.
- The result is not evidence for modified gravity or dark-energy dynamics.

10 Reproducibility

The numerical pipeline is contained in the public repository:

https://github.com/Chris4081/structural-selection-principle

The experiment folder is:

`experiments/maat_linear_perturbations_paper43/`

Run:

```
python3 maat_linear_growth_solver_v01.py
```

The script generates:

- `outputs_paper43/paper43_summary.json`,
- `outputs_paper43/paper43_growth_curves.csv`,
- `outputs_paper43/paper43_eta_scan.csv`,
- `outputs_paper43/paper43_fsigma8_comparison.csv`,
- `outputs_paper43/fig1_growth_perturbation_summary.png`,
- `outputs_paper43/fig2_eta_scan.png`.

The code uses fixed numerical settings and does not perform stochastic sampling.

11 Data Attribution and License Notes

The Planck-normalised reference parameters and compact $f\sigma_8$ comparison points are external scientific data and should be cited to the original publications/collaborations when reused or discussed. The CSV tables and figures generated here are derived analysis artifacts for reproducibility of the MAAT diagnostic calculation. No endorsement by the Planck Collaboration, survey collaborations, or original data authors is implied.

The repository code is released under the repository license.

12 Conclusion

Paper 43 introduces the first explicit linear perturbation equation in the MAAT structural-selection cosmology sequence. A bounded projection-derived structural kernel is inserted into the sub-horizon matter-growth equation through an effective coupling $\mu(z) = 1 + \eta\widehat{\mathcal{C}}_{\mathrm{proj}}(z)$.

The benchmark result is deliberately modest: the compact growth comparison prefers $\eta = 0$, while nonzero perturbative branches remain stable and produce only small, structured deviations. This establishes a controlled bridge from projection observables to growth dynamics.

The next hard step is no longer a proxy calculation, but a full perturbation implementation with metric perturbations, survey covariance, and Boltzmann-code-level comparison to cosmological data.

References

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